11kV FROM HV SWITCHBOARD
F16-SW-011
2000-F16-SLD-EL-10001

F16-SR-002
415V FLOTATION SWITCHROOM

F16-TR-008
TRANSFORMER ROOM

F16-TR-008
11/0.433kV
630kVA
Dyn11
z=4%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
MCC EARTH BAR
NEUTRAL BAR

1000/ 1A
2.5VA
5P20

F16-SW-008
THICKENER EMERGENCY POWER 415V MOTOR CONTROL CENTRE

NOTE 1
1/700A
1000/ 5A
2.5VA
5P20
1000/ 5A
2.5VA
C10.5
00F-01
TEST
PM
20A
63A
2A
63A
AFD
MCCB
1000A
50
51
51G
50G
00F-07
125A
160A
SD
NOTE 6
INC
CONTINUED ON SHT 2
2000-F16-SLD-EL-10221

415V 3NE~ 50Hz 960A 65kA for 1sec

01F-09	01R-01	01F-06	01F-01		01R-17	01R-08	01F-13	01R-04	00R-01	01F-17	01F-21								
20A	20A	20A	160A		250A	100A	20A	100A	630A	20A	20A								
2A 32A																			
70A	70A	70A	90A		90A	70A	70A	100/1A	400A	70A	70A								
			160A 160A		250A 250A				315A 315A										
EOL	EOL	EOL				EOL	EOL			EOL	EOL								
F17TH401-JB03	F17TH401-JB03	F17TH401-JB03				F17TH401-JB03	F17TH401-JB03												
JB	JB	JB				JB	JB												
M 3~						M 3~	M 3~	M 3~	CP	M 3~	M 3~								
1.1kW	1.5kW	1.5kW				30kW	2.2kW	37kW	308.6kW	0.18kW	0.18kW								
F17-FN-492 FLOTATION TAILINGS THICKENER RAKE DRIVE OIL COOLING FAN 2000-F17-SCM-EL-10209	F17-HT-539 FLOTATION TAILINGS THICKENER RAKE DRIVE OIL TANK HEATER 2000-F17-SCM-EL-10206	F17-HT-540 FLOTATION TAILINGS THICKENER RAKE LIFT OIL TANK HEATER 2000-F17-SCM-EL-10212				F17-PO-408 FLOTATION TAILINGS THICKENER RAKE DRIVE PUMP 2000-F17-SCM-EL-10210	F17-PO-409 FLOTATION TAILINGS THICKENER RAKE LIFT PUMP 2000-F17-SCM-EL-10211	F17-PP-459 FLOTATION TAILINGS THICKENER P-DUC PUMP 2000-F17-SCM-EL-10203	F24-CM-403 PLANT AIR COMPRESSOR 3 2000-F24-SCM-EL-10603	F24-FN-417 HP PROCESS WATER PUMP 1 GEARBOX FAN 2000-F24-SCM-EL-10203	F24-FN-418 HP PROCESS WATER PUMP 2 GEARBOX FAN 2000-F24-SCM-EL-10204								

F16-SR-001
11KV FLOTATION SWITCHROOM

69kW	BC 9kW	69kW
F16-DB-101 FLOTATION 11KV SWITCHROOM HV VSD AUXILIARY SUPPLIES DB 2000-F16-SCM-EL-10213	F16-BC-001 FLOTATION 11KV SWITCHROOM 48V CHARGER 2000-F16-TLD-EL-10051	F16-DB-001 FLOTATION 11kV SWITCHROOM L&SP DB 2000-F16-SCM-EL-10212

E-SHUT
F16SR002-LCS
NOTE 12

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-008 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- BIMETALLIC TEMPERATURE SWITCH
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA / xxxA — ST TRIP SET POINT / CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F16-SLD-EL-10221_Rev3

Rougher and scavenger flotation F16-SW-008 415V emergency MCC single line diagram sheet 1